

FERDER FYR, Norway

WMO: 014820

Lat: 59.027N

Lon: 10.524E

Elev: 25

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	16.9	20.0	10.3	9.3	21.3	13.7	11.1	24.1	38.9	36.5	36.7	35.4	16.1	30	0.777

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.2	73.7	66.0	71.5	64.7	69.6	63.4	67.4	72.2	65.8	70.5	64.2	68.9	9.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.4	93.9	70.8	63.7	88.3	69.5	62.0	83.0	67.7	31.8	72.5	30.6	70.6	29.4	69.1	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
37.4	34.0	31.1	DB	15.2	76.6	4.6	3.1	11.9	78.9	9.2	80.7	6.6	82.4	3.3	84.7
			WB	14.3	68.5	4.2	2.3	11.3	70.2	8.8	71.5	6.5	72.8	3.5	74.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.4	33.4	32.7	35.8	43.0	51.9	59.0	64.3	63.8	57.5	49.0	41.4	36.4
	DBStd	12.45	6.50	6.49	4.91	4.35	5.03	3.83	3.87	3.84	3.77	4.80	5.96	7.22
	HDD50	2449	516	486	439	214	34	0	0	0	1	76	260	423
	HDD65	6491	981	906	904	659	408	185	61	70	226	496	707	888
	CDD50	1514	0	0	0	5	93	271	444	428	225	45	3	0
	CDD65	78	0	0	0	0	1	5	39	33	0	0	0	0
	CDH74	51	0	0	0	0	1	3	37	10	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	16.0	18.2	17.1	15.4	13.7	14.1	14.0	13.5	14.4	16.6	17.8	19.2	18.2
Precipitation	PrecAvg	31.6	2.8	1.8	1.5	2.2	1.8	2.5	2.4	3.1	3.1	3.9	3.7	2.7
	PrecMax	48.5	5.4	4.8	5.3	4.6	3.1	5.1	5.1	6.0	5.9	9.2	13.6	4.9
	PrecMin	23.8	0.4	0.7	0.0	0.4	0.1	0.2	0.3	1.0	1.1	1.3	0.7	0.4
	PrecStd	6.4	1.3	1.1	1.3	1.1	1.0	1.2	1.1	1.5	1.4	1.9	3.0	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.7	44.9	46.3	57.3	70.1	72.8	77.6	75.4	67.4	59.4	54.7	49.9
		MCWB	43.7	40.8	41.1	50.3	60.5	62.5	67.0	66.8	60.8	55.2	51.9	47.2
	2%	DB	44.6	43.0	44.4	53.8	64.3	69.3	74.6	73.3	65.2	57.7	52.6	48.4
		MCWB	42.0	40.4	40.0	48.3	56.7	60.2	66.3	66.3	59.9	54.3	49.9	46.0
	5%	DB	43.1	41.8	42.9	51.3	61.0	66.6	72.0	71.4	63.8	56.6	51.1	47.1
		MCWB	40.6	39.7	39.6	46.7	54.8	59.0	65.0	65.3	59.1	53.5	48.3	44.6
	10%	DB	41.6	40.7	41.7	48.9	58.5	64.5	70.0	69.6	62.6	55.6	49.6	45.5
		MCWB	39.4	38.9	39.0	45.0	53.2	57.9	63.8	63.9	58.3	52.6	46.8	43.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.6	42.2	42.7	51.5	61.5	65.0	69.1	69.3	63.5	56.6	52.8	48.2
		MCDB	46.2	43.4	44.8	55.4	68.6	71.1	74.9	73.1	66.3	58.0	54.1	49.4
	2%	WB	42.6	41.1	41.4	49.1	58.2	62.1	67.5	67.8	61.5	55.4	50.5	46.5
		MCDB	44.1	42.3	43.2	53.0	62.9	67.1	72.9	71.9	64.0	57.1	52.2	48.2
	5%	WB	41.1	40.2	40.5	47.2	55.8	60.4	66.1	66.2	60.2	54.4	48.8	45.0
		MCDB	42.7	41.6	42.2	50.5	60.0	65.2	71.2	70.5	63.0	56.2	50.7	46.7
	10%	WB	39.7	39.1	39.5	45.6	53.8	59.0	64.6	64.6	59.0	53.1	47.2	43.5
		MCDB	41.3	40.7	41.2	48.4	57.6	63.4	69.3	69.2	62.1	55.2	49.2	45.3
Mean Daily Temperature Range	5% DB	MDBR	5.2	4.8	5.2	6.0	6.7	6.5	6.2	5.8	5.2	5.0	4.9	5.2
		MCDBR	5.4	4.5	5.9	8.6	9.6	8.7	8.2	7.1	5.8	4.8	5.3	4.6
		MCWBR	5.9	4.3	4.6	5.6	5.8	6.1	5.2	5.0	5.2	5.2	6.4	5.4
	5% WB	MCDBR	5.2	4.1	5.3	7.5	8.3	7.5	7.5	6.5	4.7	4.2	4.9	4.6
		MCWBR	5.7	4.4	4.4	5.3	5.5	5.5	5.0	5.2	5.0	5.1	6.2	5.4
Clear-Sky Solar Irradiance	taub	0.289	0.324	0.341	0.380	0.373	0.360	0.382	0.379	0.349	0.337	0.304	0.265	
	taud	2.246	2.326	2.370	2.315	2.413	2.474	2.433	2.436	2.503	2.428	2.290	2.170	
	Ebn at Noon	166	220	253	261	272	278	268	260	250	214	158	130	
	Edh at Noon	14	22	27	34	33	32	32	30	24	19	13	10	
All-Sky Solar Radiation	RadAvg	100	284	759	1230	1646	1872	1691	1326	881	392	133	68	
	RadStd	19	52	77	166	155	117	188	88	93	65	31	7	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.68	N/A	+2.18	N/A	N/A	-1.69	N/A	N/A	N/A	N/A
(47)	Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page